UNIX, LINUX & ANDROID

Case Study 1: The Evolution of Modern Operating Systems

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EVOLUTION ROADMAP

1969

UNIX

Born at Bell Labs. The start of it all.

1991

LINUX

The Open Source revolution by Linus Torvalds.

2008

ANDROID

The Mobile Era begins with Version 1.0.

CHAPTER 1: UNIX

The Great Origin & Portability Pioneer

THE VISIONARIES

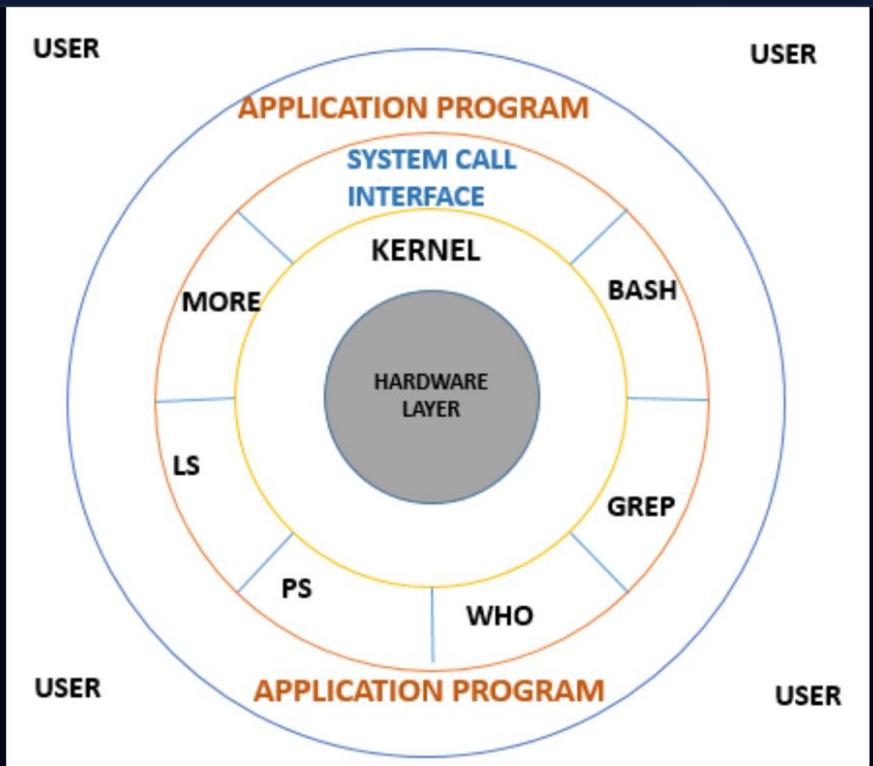
Ken Thompson & Dennis Ritchie

- Developed at **Bell Labs (1969)**.
- Rewritten in **C Language** for groundbreaking portability across hardware.
- Philosophy: **"Everything is a file"**.
- Introduced the concept of the **Shell** for user interaction.

CORE ARCHITECTURE

System Stratification

- **Kernel:** Manages CPU, RAM, and I/O directly.
- **Shell:** The command interpreter bridging User and Kernel.
- **User Space:** Where applications and utilities execute.
- **Hardware:** The foundational physical layer.



UNIX ARCHITECTURE

CHAPTER 2: LINUX

Open Source & The Monolithic Standard

THE OPEN REVOLUTION

Linus Torvalds (1991)

- Initially based on **Minix** (a UNIX variant).
- Fully **Open Source** under the GPL License.
- Not UNIX by code, but strictly **POSIX compliant**.
- Allowed a global community to contribute to the Kernel.

MONOLITHIC KERNEL STRUCTURE

⚡ Advantages

Extreme high performance as all services (VFS, IPC, Memory Management) run in Kernel Space, minimizing context switching costs.

⚠ Risks

Reduced modularity; a single driver bug or subsystem error can lead to a total system crash (Kernel Panic).

CHAPTER 3: ANDROID

Mobile Evolution on a Linux Foundation

MOBILE FOUNDATION

Android is not a traditional Linux distribution like Ubuntu, but it "stands on the shoulders" of the Linux Kernel.

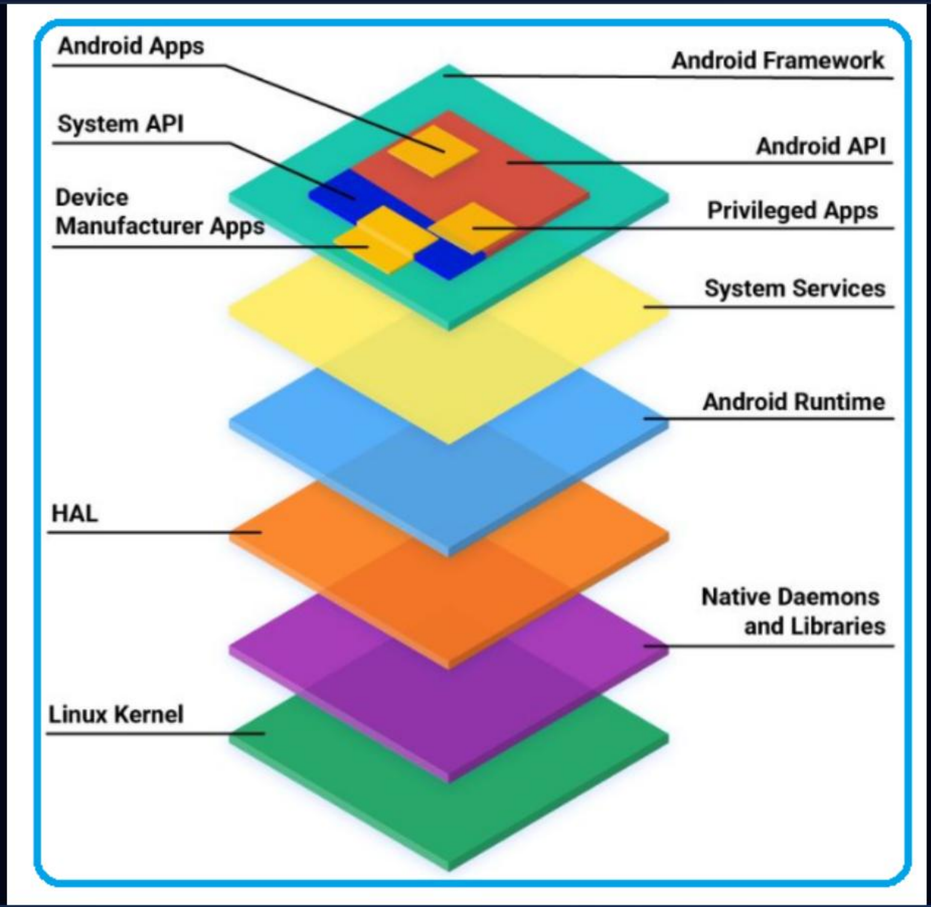
It utilizes Linux for:

- **Driver Management** for hardware components.
- **Power Management** for battery optimization.
- **Security** based on UNIX-style permissions.

THE ANDROID STACK

Architectural Layers

- **Linux Kernel:** Hardware & resource core.
- **HAL:** Standard interface for hardware.
- **ART:** Uses Ahead-of-Time (AOT) for speed.
- **Java API:** Developers' playground.
- **System Apps:** Dialers, Email, etc.



TECHNICAL COMPARISON

FEATURE	UNIX	LINUX	ANDROID
Kernel Type	Monolithic / Micro	Monolithic	Linux Kernel + HAL
Target Device	Servers / Workstations	Servers / PCs	Mobile Devices
App Execution	Native Binaries	Native Binaries	ART (Bytecode)
Source Code	Mostly Proprietary	Open (GPL)	Open (AOSP)

SYSTEM CALL PROCESS

1. Setup

App places arguments into registers or the stack.

2. Trap

Executes `trap` to switch from User to Kernel Mode.

3. Handler

OS Kernel dispatches to the correct internal handler.

4. Return

Completion, returns to User Mode and hands back control.

LEGACY & INNOVATION

55+

Years of Evolution

The Path Forward

The core principles of UNIX continue to power the digital world through Linux and Android. Stability, modularity, and community-driven innovation are the foundations of the future.

"The future of OS is distributed, secure, and mobile-first."

Q&A

Thank you for your attention!

```
root@osg202-group: help --questions
```

IMAGE SOURCES

https://upload.wikimedia.org/wikipedia/commons/1/1b/Ken_Thompson_and_Dennis_Ritchie--1973.jpg

Source: fr.wikipedia.org

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https://homework.study.com/cimages/multimages/16/unix4736320062894985953.png

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